WILLIAM LEE

Senior Platform Engineer - Kubernetes and Multi-Cloud

+1 (786) 949-7561 | williamlprodev@outlook.com | linkedin.com/in/william-lee-56a034416 | Tennessee, USA

PROFESSIONAL SUMMARY

Senior Platform Engineer - Kubernetes and Multi-Cloud with 10+ years delivering secure software across application security, device protection, learning technology, and digital media, using Kubernetes 1.18-1.31, Terraform 0.12-1.9 and modern GitOps tooling; combines hands-on architecture, modernization, incident leadership, and mentoring to build scalable systems that improve performance, reliability, security, quality, customer outcomes, and release speed.

SKILLS

Cloud & Infrastructure: AWS, Azure and GCP, networking, IAM, load balancing, DNS, certificates, secrets, managed databases, storage and serverless services

Containers & IaC: Kubernetes, Docker, Terraform, Helm, Ansible, Packer, Crossplane, operators, policy-as-code

CI/CD & Platform: GitHub Actions, GitLab CI, Jenkins, Azure DevOps, Argo CD, Flux, artifact registries, golden paths, Backstage

Observability & Reliability: OpenTelemetry, Prometheus, Grafana, Loki, ELK/OpenSearch, distributed tracing, SLOs, error budgets, capacity planning

Security & Operations: zero-trust networking, least privilege, vulnerability management, patching, backup/restore, disaster recovery, incident command

Architecture & Leadership: platform engineering, multi-region design, GitOps, immutable infrastructure, Architecture reviews, technical discovery, estimation, mentoring, code review, incident leadership, Agile delivery, stakeholder communication, roadmap planning

WORK EXPERIENCE

Self-Employed / Independent Consulting | Jun 2026 - Present | Tennessee, United States

Staff Software Engineer

- Architected and delivered **self-service developer platforms, Kubernetes clusters and reusable delivery templates** using **Kubernetes 1.28-1.31, Terraform 1.5-1.9, GitHub Actions, Argo CD and OpenTelemetry**, establishing modular boundaries, API contracts, secure defaults, and an incremental roadmap that balanced near-term releases with long-term maintainability.
- Modernized legacy services and user flows into **platform engineering, Kubernetes operators, golden paths and policy-as-code**, resolving memory leaks, N+1 database queries, race conditions, stale-cache defects, and timeout failures to reduce **p95 response time by 38%**.
- Built automated delivery on **AWS, Azure and GCP** with containers, infrastructure as code, environment promotion, secrets management, and rollback controls, shortening **release lead time by 42%** without weakening auditability.
- Designed secure **REST, GraphQL, gRPC, webhook, and event-driven integrations** with OAuth 2.0, OpenID Connect, short-lived credentials, idempotency, rate limiting, and contract-based compatibility controls.
- Introduced unit, integration, contract, end-to-end, performance, and security testing around **Kubernetes, Terraform, Backstage and multi-cloud platforms**, reducing escaped defects by **31%** and making production releases more predictable across remote teams.
- Optimized relational queries, cache strategy, search indexes, asynchronous workloads, and cloud resource sizing, cutting recurring infrastructure cost by **24%** while preserving availability and growth headroom.
- Led architecture reviews, technical discovery, estimation, code reviews, and incident retrospectives, mentoring engineers through practical design decisions rather than relying on framework-first or rewrite-first solutions.
- Implemented observability with structured logs, metrics, distributed traces, SLO dashboards, and actionable alerts, improving fault isolation and supporting a sustained **99.95% service availability** target.
- Partnered with product, security, data, and operations stakeholders to convert ambiguous requirements into sequenced technical outcomes, documented ADRs, migration plans, and measurable business acceptance criteria.

Veracode | Jul 2021 - May 2026 | Somerville, MA

Staff Software Engineer

- Led architecture and hands-on implementation for **internal engineering platforms for secure delivery, service ownership and developer productivity** using **Kubernetes 1.20-1.29, Terraform 0.14-1.7, Helm, Jenkins/GitHub Actions and Prometheus**, supporting multi-tenant SaaS growth while maintaining tenant isolation, secure coding controls, and backward-compatible customer workflows.
- Refactored tightly coupled modules into **platform engineering, Kubernetes operators, golden paths and policy-as-code**, eliminating duplicate processing, connection-pool exhaustion, queue backlogs, and inconsistent retries to improve sustained processing throughput by **34%**.
- Designed versioned APIs and asynchronous contracts for findings, policies, scans, notifications, and external integrations, applying idempotency keys, schema evolution, circuit breakers, and dead-letter recovery.
- Modernized legacy components through staged migrations, compatibility adapters, feature flags, and dual-run validation, avoiding a disruptive rewrite while steadily retiring unsupported libraries and build tooling.
- Strengthened quality gates with static analysis, dependency scanning, unit and integration suites, consumer-driven contract tests, and targeted load tests, reducing escaped production defects by **29%**.
- Improved deployment automation across **AWS, Azure and GCP** with immutable artifacts, environment policy checks, progressive rollout, and automated rollback, reducing median change lead time by **40%**.
- Diagnosed high-severity production issues involving malformed scan payloads, cross-service timeouts, duplicate events, authorization edge cases, and noisy alerts, then converted findings into durable platform fixes and runbooks.
- Mentored senior and mid-level engineers, facilitated design reviews, and aligned product, security, support, and data teams on technical tradeoffs, service ownership, operational readiness, and measurable delivery outcomes.

Asurion | May 2018 - Jun 2021 | Nashville, TN

Senior Software Engineer

- Developed and evolved **shared service templates, CI/CD foundations and developer tooling for consumer product teams** with **Kubernetes 1.12-1.21, Terraform 0.11-0.14, Jenkins and managed cloud services**, designing clear service boundaries and reusable components for customer journeys that had to remain responsive during seasonal and partner-driven traffic spikes.
- Reworked synchronous claim and diagnostic paths into asynchronous, retry-safe workflows with caching and database indexing, reducing average transaction processing time by **27%** and improving peak-load stability.
- Resolved recurring production failures caused by duplicate callbacks, partial partner responses, stale device state, slow queries, and inconsistent error mapping, adding idempotency, reconciliation jobs, and clearer operational telemetry.
- Introduced API versioning, schema validation, OAuth-based authorization, secure secret handling, and contract tests for internal and partner integrations, reducing integration regressions and support escalation volume.
- Expanded automated test coverage across unit, integration, browser or device, and performance scenarios, using CI quality gates and representative test data to reduce customer-reported software defects by **22%**.
- Supported legacy framework and runtime upgrades through dependency audits, compatibility layers, canary releases, and rollback plans, enabling modernization without interrupting claims, diagnostics, or fulfillment operations.
- Collaborated with product, UX, operations, and partner teams in Agile delivery, reviewed designs and pull requests, coached engineers, and translated customer-impacting incidents into prioritized reliability improvements.

Rustici Software | Feb 2016 - Mar 2018 | Nashville, TN

Software Engineer

- Built and maintained **reusable integration tooling, service templates and deployment foundations for learning products** using **Docker 1.10-17.x, Jenkins 2, Ansible 2 and early Kubernetes adoption**, translating SCORM, xAPI, cmi5, and LMS interoperability requirements into maintainable application modules and testable integration contracts.
- Diagnosed content-launch failures involving malformed manifests, browser security restrictions, session expiration, inconsistent completion state, and vendor-specific payloads, adding validation, compatibility handling, and clearer diagnostics.
- Improved reporting and content-processing performance through query tuning, batch pagination, background workers, cache controls, and selective denormalization while preserving standards compliance and customer data integrity.
- Modernized legacy modules incrementally with automated regression tests, dependency updates, API adapters, and documented migration paths, avoiding broad rewrites that would have increased customer integration risk.
- Implemented REST integrations, webhooks, message-based processing, authentication controls, and structured logging for customer-hosted and SaaS deployments, improving supportability across varied enterprise environments.
- Worked closely with product specialists, support engineers, and customer developers to reproduce difficult interoperability defects, review code, document behavior, and deliver fixes with clear upgrade guidance.

Mediacube | Jun 2015 - Jan 2016 | Knoxville, TN

Junior Software Engineer

- Implemented features for **shared deployment scripts, application templates and internal developer utilities** using **Linux, Bash, Jenkins 1.x, configuration scripts and virtual-machine based deployments**, converting product requirements into maintainable application code, SQL changes, background jobs, and user-facing workflows under senior engineering guidance.
- Fixed defects involving duplicate campaign records, failed media uploads, incorrect time-zone calculations, inconsistent totals, and slow reporting queries, adding validation, indexes, transaction boundaries, and actionable logging.
- Created and consumed REST endpoints, integrated third-party media and reporting services, and added retry and error-handling behavior for unreliable external responses and scheduled processing jobs.
- Expanded unit and integration checks, participated in code reviews, documented deployment steps, and supported Linux-based production releases while learning disciplined debugging and change-management practices.
- Collaborated with designers, account teams, and experienced engineers to break work into testable increments, clarify edge cases, and deliver digital-media features without disrupting existing customer reporting.

SELECTED PROJECTS

Self-Service Kubernetes Delivery Platform

Built a reference internal platform with **Kubernetes, Terraform, Helm, GitOps, policy-as-code, secrets management, service templates, observability, and automated environment provisioning** across multi-cloud services.

SLO-Driven Reliability Automation

Implemented OpenTelemetry-based tracing, service-level objectives, error-budget reporting, capacity dashboards, synthetic checks, incident runbooks, and automated remediation workflows for multi-service production environments.

EDUCATION

University of Tennessee, Knoxville | Sep 2011 - May 2013

Bachelor of Science, Computer Science